

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester II Examination 2024-25 (NEP)

DEPARTMENT OF PHYSICS**Mid-Semester Examination****Paper Code: PHYMJ21 / PHYMN21 — Thermal Physics**

Time Allowed: 1:00 Hour

Maximum Marks: 20

*Instruction: Attempt all questions.***Question 1**

[2]

Calculate the work done by 2 moles of an ideal gas in terms of gas constant R whose volume expands isothermally to 4 times of its initial volume.

Question 2 Write short notes on any two of the following:

[2.5 + 2.5 = 5]

- (i) Thermodynamic equilibrium
- (ii) Second law of thermodynamics
- (iii) Refrigerator

Question 3

[3]

Two Carnot engines X and Y are operated in series. Engine X receives heat from a source at temperature 900 K and rejects heat to a sink at temperature T K. The second engine Y receives heat at temperature T K and rejects heat to its sink at temperature 400 K. For what value of temperature T will the efficiencies of both engines be the same?

Question 4

[5]

Write Maxwell's law of speed distribution and by using this derive the relation for the most probable speed of gas molecules.

Question 5

[2]

Write the relationship between mean free path and gas temperature. Explain the terms used.

Question 6

[3]

If the diameter of a gas molecule is 2×10^{-8} cm, find the value of mean free path at STP ($T = 273$ K, $P = 1.013 \times 10^5$ N/m², $k = 1.38 \times 10^{-23}$ J/K).

[Click here to visit our website](#)

[Click here to visit our contributor](#)